BT601 (Virology)

Current + Past Questions Solved

Midterm Spring – 2023

We Grow Together

Solved By Bella

Dyes used in TEM? 3 Marks

Uranyl acetate, lead citrate, and osmium tetroxide are commonly used dyes in Transmission Electron Microscopy (TEM) to enhance contrast and visualize biological specimens.

OR

Contrast can be greatly enhanced by using a "dye" that absorbs electrons and produces a darker image in the stained region lead citrate, osmium tetroxide, tungsten, uranyl acetate and uranium (Stains). These metals can be fixed onto the specimen (positive staining) or used to increase the electron opacity of the surrounding field (negative staining).

Principle of fluorescence antibody technique? 3 Marks

Based on the observation that virus proteins are (antigen) present in the tissues e.g rabies virus present in nervous tissues. The principle of the Fluorescence Antibody Technique (FAT) is based on the use of fluorescent-labelled antibodies that specifically bind to the antigen of interest. The fluorescent tag allows for the visualization of the bound antibody-antigen complex, which is detected using a fluorescence microscope. The technique is highly specific and sensitive, allowing for the rapid and accurate detection of a wide range of pathogens in biological samples.

Effects of physical and chemical factors on viruses?

Inactivation of virus represents permanent loss of contamination.

Temperature: viruses are stable on low temperature (keeping) viruses with envelope are heat sensitive

Drying: most of the viruses are sensitive Irradiation (UV, X-ray) inactivation of viruses.

Chemical agents

Organic solvent (chloroform) – inactivation of the viruses with envelope oxido-reduct. Agents (formaldehyde, chlorine, iodine) – inactivation.

pH low pH – viruses without envelope are mostly stable

Effect of viral attack on host cell. 3 Marks

When viruses attack host cells, they replicate inside the host to produce new viral particles and use the host's metabolism, which can damage or destroy the host cell. The process can trigger host defence mechanisms that may cause tissue damage, contributing to the symptoms of the disease. Consequently, the damage can cause pathological effects, which could result in inflammation, tissue damage, cell death, and organ dysfunction, having grave implications for the host's health.

What is complementation? Also write its types.

Assignment of mutants to functional groups resulting production of one or both parent viruses. One of the viruses provides functional gene product allows functional analysis of unknown mutations.

Types of complementation:

Allelic (intragenic) complementation where different mutants have defects in same protein.

Non-allelic (intergenic) results from mutants with defects in different genes.

How viral suppression is done? Or Define informational suppression?

Mechanism different from suppression of chain-terminating amber mutations by host encoded suppressor tRNAs that is also referred as informational suppression.

Write down the ways of reversion?

Mutant viruses can revert by three ways:

True reversion

- Back mutation of original mutation
- Gives wild-type genotype or phenotype

Intragenic reversion

- Compensatory mutation, which occurs in same originally mutated gene
- Restoring it

Extragenic Reversion

- A suppressor mutation in a different gene
- Virus gene or host gene

What is suppression of a viral genome?

Inhibition of a mutant phenotype by a second suppressor mutation which may be either in viral genome or in host cell.

What is viral recombination and intramolecular recombination?

Viral recombination

The exchange or transfer of genetic material between different but closely related viruses infecting the same cell.

Intramolecular recombination

- Usually in DNA viruses
- Dissociation and re-establishment of covalent bonds with nucleic acids.
- Template switching mechanism

Define genetic reactivation?

Infectious progeny produced from parents, in which one or both are non-infectious mixed infection of cell.

Define following terms?

Mutation:

- A permanent inheritable change in a single gene that results in the existence of two or more alleles occurring at the same locus.

Pseudotype:

The viruses that have acquired the genotype of one parent and phenotype of another.

Recombinant:

- A new combination of genes not found together on the chromosome in either parent.
- A recombinant results from crossing over.

Codon:

- A sequence of three bases in DNA or RNA that codes for single amino acid
- Enables specific proteins to be made by specific genes.

Anti-Codon:

- A sequence of three bases in tRNA that is complementary to a codon in mRNA
- Enables tRNA to sequence amino acids in the order specified by mRNA.

Ribosomes:

- Complexes of rRNA and protein in cytoplasm that serve as platforms for translation for mRNA into protein.

Plasmid:

- Extrachromosomal circular DNA in bacteria
- Independently replicate and encode a product

Transcription:

- Synthesis of single-stranded RNA by RNA polymerase using DNA as a template
- The process in the nucleus whereby DNA is transcribe into mRNA.

Translation:

- The process of translating the codon sequence in mRNA into polypeptides with the help of tRNA and ribosomes.

Write down the names of the negative Sense RNA Plant Virus?

- Bornaviridae
- Bunyaviridae
- Orthoviridae
- Paraviridae
- Rhabdoviridae

Write down the names of the positive Sense RNA Plant Virus?

- Luteviridae
- Sobemovirus
- Sequiviridae
- Sadwavirus
- Chevavirus
- Tombusviridae
- Tymoviridae
- Comoviridae
  Idaeovirus
- Bromoviridae
- Cucumovirus
- Bromovirus
- Ilavirus
- Alfamovirus
- Ourmiavirus
- Tobamovirus
- Hordievirus
- Pecluvirus
- Furovirus
- Pomovirus
- Bemivirus
- Flexiviridae
- Potyviridae
- Closteroviridae

Write down the classification of Bunyaviridae?

Largest family of 5 genera:

1. Orthobunyavirus
2. Hantavirus
3. Nairovirus
4. Phleovirus
5. Tospovirus

What are latent occult infections of viruses.

Some viruses reappear to cause acute disease

Three examples

- HSV
- Adenovirus
- SSPE

HSV

- Herpes Simplex Virus
- Primary infection in early age
- Found only during recurrent acute episodes
- Virus can be isolated from tissue homogenates

ADENOVIRUS

- Self-limiting
- Persistent infection of tonsils and adenoids
- Failure in viral isolation
- Absence of mature virion
- DNA in linear episomal form

MEASLES VIRUS

- Incomplete viral production
- Immature nucleo-capsids produced
- Disease depends on immune response
- May cause tumorigenesis

Major vectors reason of viruses in plants?

1. Insect vectors (aphids, whiteflies, thrips, and leafhoppers)

2. Fungi

3. Nematodes

4. Mites

5. Environmental conditions (wind, rain, irrigation, and equipment movement)

Fluorescence antibody technique? 5 Marks

The Fluorescence Antibody Technique (FAT) is a widely used immunological method for detecting and identifying pathogens based on the specific binding of fluorescent-labelled antibodies to their antigens.

- The FAT involves the use of fluorescent-labelled antibodies that specifically bind to the antigen of interest, resulting in a fluorescent signal that can be visualized under a fluorescence microscope.
- The technique can be used to detect a wide range of antigens, including viral, bacterial, fungal, and parasitic antigens, as well as tumor markers.
- The technique is highly specific and sensitive, allowing detection of very low levels of pathogen antigens in biological samples.
- The FAT is widely used in clinical microbiology and virology for the rapid and accurate diagnosis of infections caused by viruses and other pathogens.
- In addition to diagnosis, the FAT is also used for research purposes, such as studying the localization and distribution of specific antigens within cells or tissues.

Protein & glycoprotein role of life cycle of virus? 5 Marks

Proteins and glycoproteins have essential roles in the life cycle of viruses. They facilitate the attachment and recognition of host cells, assist in the penetration of viral genome into the host cell, and interact with the host cell machinery to initiate the replication and transcription of viral genomes. Furthermore, the assembly of new viral particles, genome packaging, and release of new viral particles is also reliant on viral proteins and glycoproteins. Proteins and glycoproteins are involved at every stage of the viral life cycle, playing crucial roles in the process.

Which type of centrifuge is best for isolation of viruses?

Ultracentrifuge

Ultracentrifuges are typically considered to be the best type of centrifuge for virus isolation as they have the capability to separate viral particles from other contaminants at high speed and with high precision. These centrifuges typically operate at speeds of up to 100,000 rotations per minute (rpm) and have capacities ranging from a few millilitres to several litres. Additionally, ultracentrifuges are capable of separating individual viral particles based on differences in size, density, and buoyant density, which can be important for the purification of virus particles from complex biological mixtures. Thus, ultracentrifuges are widely accepted as the gold standard for virus isolation and purification.

What is lysogenic part of lifecycle?

The pathogenic viruses having the ability to lie dormant (or latent) within host cell called lysogenic part of lifecycle.

Define latency?

- The phase in which viral proliferation ceases after initial infection.
- Viral genome not fully eradicated.

Host range of virus?

- The spectrum of host cells that the virus can infect.
- Host range refers to the spectrum of host cells in which a virus can multiply.
- Most viruses infect only specific types of cells in one host species.
- Host range is determined by the specific attachment site on the host cell's surface and the availability of host cellular factors.
- Most viruses are host specific bind to specific receptors on cells
- Animal viruses: receptors on plasma membrane form hydrogen bond with viral surface

Viral host range

- Plants
- Vertebrates
- Invertebrates
- Protists
- Bacteria
- Fungi

Genetic makeup of retrovirus?

- Diploid nature of virus.

3 very important genes:

GAG

- Gag gene makes protein p24 which serves as capsule around RNA and in cone shaped.

ENV

- Envelope glycoproteins (gp41 and gp120)
- Gp41: transmembrane protein
- Gp120: outer glycoprotein
- Host receptor

POL

- Reverse transcriptase polymerase
- Integrase activity
- DNA from RNA

Name 2 Type of mutations that can be induced . Write mutagen of each category.

IN-VITRO

- Chemically modify nucleic acids
- Do not require replication
- E.g: nitrous acid, hydroxylamine and alkylating agents like nitrosoguanidine

IN-VIVO

- Requires Metabolically active nucleic acid that is replicating
- Compounds incorporated into newly synthesized nucleic acids
- E.g:5-bromouracil resulting erroneous base pairing, intercalating bases that stack between bases, UV irradiation

Write two limitations of electron microscope .

Only a very thin section of a specimen (about 100 nm) can be studied effectively. Specimens must be fixed, dehydrated, and viewed under a high vacuum to prevent electron scattering. These treatments not only kill the specimen, but also cause shrinkage and distortion.

MUTATION: GOOD OR BAD?

Sometimes good for the virus and helps escape the immune system but some mutations may be deleterious due to production of defective particles.

DO RNA VIRUSES HAVE PROOF-READING?

Yes, Yet they show higher rates of mutation than DNA viruses.

Name of in vitro genetic techniques in virology.

In-vitro

- Physical analysis
- Isolation of nucleic acids from virus preparations
- Direct analysis of DNA using restriction endonucleases enzyme
- Use of vectors in DNA sequencing
- Electron microscopy
- RNA sequence analysis using reverse transcriptase

In-vivo (viral genotype and phenotype)

Molecular Analysis

- Molecular cloning
- Gel electrophoresis
- Hybridization of complementary nucleotide sequences

Phylogenetic Analysis

- Standard technique in virology
- Cell-free extracts to translate mRNAs
- Transfection tests
- In-vitro protein formation
- In-vivo protein replication

Different the principles of Scanning Electron Microscopy and transmission Electron microscopy? 5 Marks

SEM

An electron gun produces a finely focused beam of electrons called the primary electron beam. These electrons pass through electromagnetic lenses and are directed over the surface of the specimen. The primary electron beam knocks electrons out of the surface of the specimen, and the secondary electrons thus produced are transmitted to an electron collector, amplified, and used to produce an image on a viewing screen or photographic plate. The image is called a scanning electron micrograph.

TEM

Focused beam of electrons from an electron gun passes through the ultrathin section of the specimen. Beam is focused on a small area of the specimen by an electromagnetic condenser lens. Instead of being placed on a glass slide, the specimen is usually placed on a copper mesh grid.

Define following terms?

Strain: lines or isolate of same virus
Type: serotypes of same virus
Variant: virus, whose phenotype varies from original virus
Mutant: virus showing genetic alteration

Define Mutant?

A mutant is an organism or a new genetic character arising or resulting from an instance of mutation, which is generally an alteration of the DNA/RNA sequence of the genome or chromosome of an organism.

Write down the types of Mutant Viruses?

- Biochemical Markers
- Deletions
- Host Range
- Non-Sense
- Plaque morphology
- Temperature sensitive
- Cold Sensitive
- Reverents

Write at least 3 roles of viral capsid? 3 Marks

- Protection of genome
- Prevent digestion by cellular enzymes
- In cytoplasm Helps to introduce genome in host cell
- Bind to cell surface

Difference between Capsid and Envelope?

Capsid

The capsid (coat) protein is the basic unit of structure; functions that may be fulfilled by the capsid protein are to:

- Virus particles contain the viral genome packaged in a protein coat called the capsid.
- Protect viral nucleic acid
- Interact specifically with the viral nucleic acid for packaging
- Interact with vector for specific transmission
- Interact with host receptors for entry to cell
- Allow for release of nucleic acid upon entry into new cell
- Assist in processes of viral and/or host gene regulation

Envelope

- Surrounding the capsid composed of lipid bilayer that contains viral proteins derived from the host cell membranes.
- In some viruses, the capsid is covered by an envelope
- Lipids, proteins, and carbohydrates
- Spikes
- Projection from the surface of the envelope
- Viruses attach to host cells
- Influenza virus to clump red blood cells is associated with spikes
- Viruses whose capsids are not covered by an envelope are known as nonenveloped viruses

Write 5 family and their genera Negative Sense RNA viruses? 5 Marks

Bornaviridae

- Bornavirus

Bunyaviridae

- Bunyavirus
- Hantavirus
- Phlebovirus

Orthoviridae

- Influenza virus A,B,C
- Thogotovirus

Paraviridae (Paramyxovirinae , Pneumovirinae)

- Respirovirus
- Morbillivirus
- Rubulavirus
- Pneumovirsu
- Metapneumovirus

Rhabdoviridae

- Lyssavirus
- Vessiculovirus
- Ephemerovirus
- Novirhabdovirus

Scanned probe microscopy? 5 Marks

Forms images of surfaces using physical probe that scans the specimen and founded in 1981.

Principle:

The interaction between the sharp probe and surface provides a 3D topographic image of the surface at the atomic scale. Two most popular modes:

- Scanning Tunnelling Microscopy (STM)
- Atomic Force Microscopy (AFM)

Image Formation:

- Scans tip over surface
- A value is recorded
- Displayed as heat map

Limitations:

- Detailed shape of probe is difficult to determine sometimes
- Slower in image acquiring
- The maximum image size is smaller
- Cannot examine buried solid-solid or liquid-liquid interfaces.

What is RNPs?

The segmented genome is encapsidated with each segment in a separate nucleocapsid called as ribonucleoprotein complexes (RNPs). Terminal sequences of RNA bound to trimeric RdRp. All these RNPs are surrounded by one envelope.

General characteristics of virus. 5

1. A virus is a microscopic infectious agent that can only replicate inside the living cells of a host organism.
2. Viruses have a simple structure, consisting of genetic material (DNA or RNA) wrapped in a protein coat called a capsid. Some viruses also have an outer envelope made up of lipids.
3. Viruses are considered to be non-living as they lack the ability to carry out metabolic functions and reproduce on their own.
4. Viruses are highly specific, targeting specific host cells based on the interaction between viral proteins and host receptors.
5. Viruses can infect a wide range of organisms, including plants, animals, and even bacteria (known as bacteriophages).
6. Viral infections can cause a wide range of diseases in their hosts, from mild illnesses to severe infections.
7. Viruses can be transmitted through various routes such as respiratory droplets, bodily fluids, contaminated surfaces, or through insect vectors.

8. Vaccines and antiviral medications have been developed to prevent and treat viral infections, but the effectiveness varies depending on the specific virus.

Mechanism of enzyme action?

Enzymes work by binding with specific substrates to form enzyme-substrate complexes. This binding process lowers the activation energy required for the reaction to take place. Enzymes may increase reaction rates by weakening bonds or providing reactive groups and surfaces that enhance the chemical interaction between substrates. The reaction is then facilitated and product formation is accelerated. Once the reaction is complete, the product is released from the enzyme, which can then be used again to catalyse further reactions. This repetitive use by enzymes makes the process highly efficient, specific, and essential for various biological processes.

Orthomyxoviridae proteins? 5 Marks

- **Hemagglutinin (HA):**

A surface glycoprotein that is responsible for virus attachment and entry into host cells.

- **Neuraminidase (NA):**

A surface glycoprotein that cleaves sialic acid from host-cell membranes, thereby facilitating the release of new virions from infected cells and the spread of the virus throughout an infected host.

- **Matrix protein (M1):**

A protein that links the viral envelope (lipid bilayer) to the viral ribonucleoproteins (RNPs) and mediates viral assembly and budding.

- **Nucleoprotein (NP):**

A protein that complexes with viral RNA to form the viral RNPs, which are responsible for viral genome replication and viral transcription.

- **RNA polymerase:**

A trimeric complex composed of three viral proteins (PB1, PB2, and PA) that catalyzes viral RNA synthesis (replication and transcription) in the host cell nucleus.

Name 3 disease cause by Peribunyaviridae? 3 Marks

1. Rift Valley fever
2. La Crosse encephalitis
3. California encephalitis

Episomal , proviral and maintaining latency? 3 Marks

EPISOMAL LATENCY

- Use of genetic episomes
- Viral genes float in cytoplasm or nucleus
- More vulnerable to host immune system
- Example, chicken pox and Herpes simplex viruses

PROVIRAL LATENCY

- Virus genome is integrated into the DNA of a host cell
- Automatic host cell division during viral replication
- Viral genome enters nucleus
- Example; HIV

MAINTAINING LATENCY

- For continuity of infection
- By genes expressed during latency
- To evade host immune system
- To protect virus genome to be digested by cellular enzymes

T phages? 3 Marks

- Head and tail called binal symmetry
- Icosahederal head and helical tail symmetry

Replication of T-Phages?

Replication Stages

1. Attachment
2. Penetration
3. Uncoating
4. Biosynthesis
5. Release

Attachment

- Specific receptors
- Techoic acid
- Lipids
- Proteins
- Flagella
- On specific bacterial hosts

Penetration

Leave capsid outside

Biosynthesis

- Bacterial ribosomes translate viral proteins
- Proteins modify bacterial RNA polymerase
- Proteins for new virions

Assembly

- T4: helper proteins required
- Base plates assembled first
- Tails built afterwards
- Heads constructed separately
- Whole process takes 15 minutes

Release

- Lysis
  - Tailed phages
  - Endolysin
- Extrusion
  - Enveloped
- Budding
- Secretion

What is reverse transcriptase?

Reverse transcription is a process in which a double standard DNA molecules ae made from a single stranded RNA. The name of this method originated by tis opposite direction to transcription process. It involves the presence of

- Reverse transcriptase, an enzyme
- A primer some bases of nucleotides segments
- Reverse transcriptase inhibitor leucine rich repeat protein

Define transposons? Also write its types and properties.

A sequence of DNA that can change its position, from place to place within a genome.

TYPES of Transposons

Two groups:

SIMPLE TRANSPOSONS

- In prokaryotes which do not undergo reverse transcription
- E.g. genome of Enterobacteria phage Mu

RETRO TRANSPOSONS

- Closely resemble retroviral genome
- Bound by long repeat units (LTRs)
- Move by means of transcription/ reverse transcription/ integration mechanism
- Found in eukaryotes
- E.g: In Metaviridae and Pseudoviridae

PROPERTIES

- Responsible for high mutation
- Promote genetic rearrangements like deletion, inversion, duplication etc
- Often accompanied by replication
- Control their own transposition functions

Agglutination and precipitation tests? 5 Marks

Agglutination

Agglutination is the process of clumping of antigens with their respective antibodies.

Precipitation

Precipitation is a process where soluble antigens bind with their specific antibody an optimum temperature and pH, resulting in the formation of an insoluble precipitate.

PRINICPILE

Viruses maybe sensitized by specific antibody. In the presence of electrolyte the virus particles cling to each other forming aggregates. Observed through examinations

- Microscopic
- Macroscopic

ANTIGEN PREPARATION

- Centrifugation
- Washing in buffered solution
- Re-centrifugation

APPLICATIONS

Two ways:

1. If virus is known, antibodies can be detected.
2. If known serum is available an unknown virus can be detected.

I/E, I/C and I/V inoculation? 3 Marks

Inoculated

- Incubated for few days
- Lesions or death observed
- These inoculations of embryos are not widely used.

Embryonic inoculation:

Embryos are inoculated and incubated for a period of one to six days during which they are observed twice a day for the evidence of death of embryo. Some viruses, such as equine encephalomylietis virus kill the embryo within 24 hours while others kill the embryos in 2 to 3 days. Some viruses readily multiply within the embryonic tissue and are seldom lethal. Embryo death is accompanied by collapse of the large blood vessels of the chorioallantois so that they are difficult to see upon candling the egg, whereas they are quite distinct while the embryos are alive. Normal movements of the embryo are no longer observed if the embryo is moribund or dead. If the embryo is not dead 5 or 6 days of incubation, it is killed or examined for lesions.

Why viruses need host?

- No enzymes of their own
- No metabolism
- Require host machinery

Role of Edward Jenner in vaccine? 5 Marks

Edward Jenner (1796) introduced the concept of vaccination. Edward Jenner was an English physician and scientist who developed the first vaccine against smallpox. He pioneered the concept of vaccination by using cowpox virus, which was a milder form of smallpox, to confer immunity to smallpox. In 1796, he tested his theory by injecting a young boy with cowpox, and this experiment was the first successful vaccine trial in history. Jenner published his findings and started advocating for the use of cowpox vaccination to prevent smallpox, which was quickly adopted throughout Europe and the world. His contribution to public health is estimated to have saved millions of lives from the ravages of smallpox and his work paved the way for the development of other vaccines. Jenner's achievement was recognized by a number of honors, including a Royal Society medal, a gold medal from Napoleon, and a knighthood from King George IV.

- Cow pox against small pox
- Adopted worldwide in 19th century
- Smallpox vaccine using milder cowpox virus
- No knowledge of viral particles

Replication of Corona virus? 5 Marks

- Attachment of the viral S protein receptors
- Endocytosis of the virus into the host cell
- Fusion of virus membrane with endosomal membrane
- Genome is released into the cytoplasm.
- Synthesis and proteolytic cleavage of the replicase polyprotein.
- A dsRNA genome is synthesized from the genomic ssRNA(+).
- The dsRNA genome is transcribed by providing viral mRNAs/new ssRNA(+) genomes.
- Synthesis of structural proteins encoded by subgenomic mRNAs.
- Assembly and budding at membranes of the endoplasmic reticulum (ER), the intermediate compartments, and/or the Golgi complex.
- Release of new virions by exocytosis.

Genome of toga virus? 3 Marks

- Linear, non-segmented, single-stranded, positive sense RNA.
- 10,000–12,000 Nucleotides
- The virus is enveloped and forms spherical particles (65–70 nm diameter)
- The capsid is icosahedral, constructed of 240 monomers, having a triangulation number of 4.

Complement Fixation Test? 5 Marks

The complement fixation test is an immunological test used to detect the presence of either specific antibody or specific antigen on the basis of complement fixation. The complement fixation is a system of serum proteins that react with antigen-antibody complexes. If this reaction occurs on a cell surface, it will result in the formation of trans-membrane pores and therefore destruction of the cell.

Short Procedure:

Serum is separated from the patient. Patients naturally have different levels of complement proteins in their serum. To negate any effects this might have on the test, the complement proteins in the patient's serum must be destroyed and replaced by a known amount of standardized complement proteins. The serum is heated in such a way that all of the complement proteins—but none of the antibodies—within it are destroyed. A known amount of standard complement proteins are added to the serum. (These proteins are frequently obtained from guinea pig serum.) The antigen of interest is added to the serum. Sheep red blood cells (sRBCs) which have been pre-bound to anti-sRBC antibodies are added to the serum.

Result Interpretation:

If the patient's serum contains antibodies against the antigen of interest, they will bind to the antigen in step 3 to form antigen-antibody complexes. The complement proteins will react with these complexes and be depleted. Thus when the sRBC-antibody complexes are added in step 4, there will be no complement left in the serum. However, if no antibodies against the antigen of interest are present, the complement will not be depleted and it will react with the sRBC-antibody complexes added in step 4, lysing the sRBCs and spilling their contents into the solution, thereby turning the solution pink

Name the 3 species of Rhabdoviridae? 3 Marks

1. Rabies virus
2. Vessicular Stomatitis Virus
3. Bovine Ephemeral fever Virus

Write the 5 types of Arboviruses?

Five viral families that comprise the majority of arboviruses transmissible to humans and livestock including:

1. Alphaviruses (Togaviridae)
2. Flaviviruses (Flaviviridae)
3. Rhabdoviruses (Rhabdoviridae)
4. Bunyaviruses (Bunyaviridae)
5. Reoviruses (Reoviridae)

Name 4 viruses of family Peribunyaviridae? 3 Marks

- California encephalitis virus
- La Crosse encephalitis virus
- Jamestown Canyon virus
- Snowshoe hare virus

Write step involve in replication of toga virus? 3 marks

1. Entry into the host cell is achieved by attachment of the viral E glycoprotein to host receptors which mediates clathrin-mediated endocytosis.
2. The receptors for binding are unknown glycoprotein petal-like spikes act as attachment proteins.
3. After virus attachment and entry into the cell, gene expression and replication takes place within the cytoplasm.

What is Toga virus? 3 Marks

- Linear, non-segmented, single-stranded, positive sense RNA.
- 10,000–12,000 Nucleotides
- The virus is enveloped and forms spherical particles (65–70 nm diameter)
- The capsid is icosahedral, constructed of 240 monomers, having a triangulation number of 4.

What is Paramyxo? 3 Marks

- Para means beyond and Myxo means mucus.
- Paramyxo and orthomyxo were formerly known as myxoviruses.

What is a reason behind the spontaneous mutations?

Spontaneous Mutation that arise naturally during replication.

REASON

- Mechanism of genome replication
- RNA dependent RNA polymerase shows higher error rate than DNA dependent DNA polymerase.

What is the mode of transmission of arena viruses? (2)

Arena viruses are a family of viruses that are usually transmitted from rodents to humans. Arena viruses are shed into the environment in the urine, saliva, or droppings of infected rodent hosts. People become infected by breathing in the virus after rodent urine, droppings or nesting materials are stirred up, such as during cleaning.

Describe the host range virus? (3)

Host ranger refers to the spectrum of host cells in which virus can multiply. Most viruses infect only specific types of cells in one host species. Host range is determined by the specific attachment site on the host cells surface and the availability of host cellular factors. Viruses that infect bacteria are called bacteriophages, or phages.

What are the 3 most efficient theories for evolution? (3)

FIRST THEORY: They may have originated and evolved with the other life forms

SECOND THEORY: they may have arisen from segments of the cellular nucleic acids which acquired the ability to replicate at the expense of host cell.

THIRD THEORY: it is Regressive theory. According to this theory, Viruses arose from free living organisms which gradually lost genetic information. Until they became totally dependent on the biosynthetic pathways of their host cells.

What are the functions of different protein of paramyxoviridae? (5)

- N – The nucleocapsid protein associates with genomic RNA and protects the RNA from nuclease digestion.
- P – The phosphoprotein binds to the N and L proteins and forms part of the RNA polymerase
- Complex
- M – The matrix protein assembles between the envelope and the nucleocapsid core; it organizes and maintains virion structure
- F – the fusion protein projects from the envelope surface as a trimer, and mediates entry.
- H/HN/G – the cell attachment proteins span the viral envelope and project from the surface as spikes.
- L – The large protein is the catalytic subunit of RNA dependant RNA polymerase (RDRP) accessory proteins – a mechanism known as RNA editing allows multiple proteins to be produced from the P gene.

Phase contrast microscope principles and it applications? (5)

Phase-contrast microscopy (PCM) is an optical microscopy technique that converts phase shifts in light passing through a transparent specimen to brightness changes in the image. Phase shifts themselves are invisible, but become visible when shown as brightness variations.

Principle:

When light passes through cells, small phase shifts occur. These phase shifts are converted into changes in amplitude, which can be observed as differences in image contrast.

Application:

- Cell culture
- Live cell imaging
- Living cells can be observed in their natural state without previous fixation or labelling.

Genome structure of retroviruses? (5)

Genome of retrovirus:

- Single stranded RNA viruses
- Positive sense
- Enveloped
- Changes to DNA by RT enzyme

There are three important genes

1. GAG
2. ENV
3. 3. POL

Structure of retroviruses:

It is the single stranded positive sense of RNA viruses. It is icosahedral in shape. All the retrovirus is surrounded by viral envelope. It contains two identical RNA and reverse transcriptase enzymes.

Latent virus types? (3)

LATENT VIRAL INFECTIONS

- Chronic persistent infections
- Latent Occult Viral infections

Differential interference contrast microscopy?

DIC also known as Nomarski interference contrast microscopy. It is used to enhance the contrast in unstained, transparent samples. DIC works on the principle of interferometry to gain information about the optical path length.

Application:

DIC is a more expensive, label-free microscopy technique with a high sensitivity to thin cellular material, even when it is located within thick tissue. It is useful for rendering contrast in transparent samples and gives brilliant pseudo-3D relief shading images.

How many types of tissue culture? 3 Marks

There are three types of tissue culture:

1) Organ culture

2) Explant culture

3) Cell culture

Replication Toga virus genome?

- Viral RNA acts as both genome as well as m RNA
- After fusion of viral capsid with cell receptor
- RNA released in cytoplasm

Viral classification of picornavirus?

A picornavirus is a virus belonging to the family Picornaviridae, and order Picornavirales vertebrates, including humans, serve as natural hosts. Birds also serve as host picornaviruses are nonenveloped viruses that represent a large family of small, cytoplasmic, plus-strand RNA (~7.5kb) viruses 30-nm icosahedral capsid

Names of 10 plant viruses having positive sense RNA? 5 Marks

- Bromoviridae
- Cucumovirus
- Bromovirus
- Ilavirus
- Alfamovirus
- Ourmiavirus
- Tobamovirus
- Hordievirus
- Pecluvirus
- Furovirus
- Pomovirus
- Bemivirus

Principle of Fluorescence microscope?

The specimen is illuminated with light of a specific wavelength which is absorbed by the fluorophores, causing them to emit light of longer wavelengths. Florescent dyes are used for specimen staining • Examples of Fluorescent stains: DAPI and Hoechst.

What is virus neutralization test?

It is a serological test that detects the presence of viral neutralizing antibodies. The antibodies bind to viral particles. These antibodies block viral infectivity and prevent from viral infections.

Baltimore classification? (5)

Baltimore system of classification 1971

Type of nucleic acid

- Mode of replication
- Host organism
- Diseases caused

Characteristics of virus? (5)

Following are the general properties of viruses

- Non living structure
- Non cellular
- Contain a protein coat known as capsid
- Contain the nucleic acid core containing DNA or RNA
- Can produce in host cells
- Some virus may have spikes to help attach to the host cells

Techniques used for the detection of viral cell genome? (5)

Classical Techniques

1. Complement fixation tests (CFT)
2. Hem-agglutination inhibition tests
3. Immunofluorescence techniques (IF)
4. Neutralization tests (Viral & Serum)
5. Single Radial Haemolysis

Newer Techniques

1. Radioimmunoassay (RIA)
2. Enzyme linked immuno-sorbent assay (EIA)
3. Particle agglutination
4. Western Blot (WB)
5. Recombinant immuno-blot assay (RIBA)

Write the name of two plants viruses (2)

1. Pepper mild mottle virus
2. Leaf curl virus.

In vitro mutation? (3)

Another use of cloned DNA is in vitro mutagenesis in which a mutation is produced in a segment of cloned DNA. The DNA is then inserted into a cell or organism, and the effects of the mutation are studied. Mutations are useful to geneticists in enabling them to investigate the components of any biological process.

Mechanism of viral attack on the host cell (5)

Viral entry pathways:

Virus can fuse either directly to the plasma membrane (receptor-mediated fusion) or after being swallowed into an endosome. Which of these routes is followed depends on the type of virus. In fusion with the plasma membrane, the virus binds to a protein in the cell membrane. The function of this cellular protein (a receptor for the virus, shown in green) is perverted to induce a conformational change in the viral fusion protein, leading to fusion. For virus that is triggered within an endosome, the endosome's acidic conditions induce fusion. In either case, the viral genome passes through a fusion pore into cytosol, and infection is initiated.

Genetic makeup of retrovirus? (10)

Diploid nature of virus 3 very important genes:

– GAG

- ENV
- POL
- GAG

Gag gene makes protein p24 serves as capsule around RNA cone shaped

ENV

- Envelope glycoproteins gp41 and gp120
- Gp41: trans membrane protein
- Gp120: outer glycoprotein
- Host receptor

POL

- Reverse transcriptase Polymerase
- Integrals activity
- DNA from RNA

Differentiate between agglutination and precipitation? (3)

Agglutination is the process of clumping of antigens with their respective antibodies.

Precipitation is a process where soluble antigens bind with their specific antibody an optimum temperature and pH, resulting in the formation of an insoluble precipitate.

What is genome?

A genome is the complete set of genetic information in an organism. It provides all of the information the organism requires to function. In living organisms, the genome is stored in long molecules of DNA called chromosomes.

Replication of toga virus? 3

Entry into the host cell is achieved by attachment of the viral E glycoprotein to host receptors which then mediates clathrinmediated endocytosis. The receptors for binding are unknown. Glycoprotein petal-like spikes act as attachment proteins. After virus attachment and entry into the cell, gene expression and replication takes place within the cytoplasm.

In-vitro technique? 3

- Physical analysis of isolation of nucleic acids from virus preparations.
- Direct analysis of DNA using restriction endonucleases enzyme.
- Use of vectors in DNA sequencing.
- Electron microscopy.
- RNA sequence analysis using Reverse transcriptase.

M13 phage? 5

M-13 PHAGE

- Simple genome
- 6.4kb
- Single stranded
- Positive sense
- Circular DNA encoding ten genes
- Icosahedral symmetry
- Capsid can be expanded by protein subunits
- Hence, genomic size can also be increased

Define “mice” as a research model?

Mice are the most commonly used animal model for studying human disease, and for many good reasons: Mice are biologically very similar to humans and get many of the same diseases for the same genetic reasons. Mice can be genetically manipulated to mimic virtually any human disease or condition.

Ribonucleic proteins?

Ribonucleoprotein (RNP): is a complex of ribonucleic acid and RNA-binding protein. These complexes play an integral part in a number of important biological functions that include transcription, translation and regulating gene expression and regulating the metabolism of RNA. A few examples of RNPs include the ribosome, the enzyme telomerase, vault ribonucleoproteins, RNase P, hnRNP and small nuclear RNPs (snRNPs), which have been implicated in pre-mRNA splicing (spliceosome) and are among the main components of the nucleolus.

What are viruses?

Viruses are complicated assemblies of molecules, including proteins, nucleic acids, lipids, and carbohydrates. Viruses can only replicate themselves by infecting a host cell and therefore cannot reproduce on their own.

Genetic techniques used in virology?

- In-vitro analysis (physical analysis)
- In-vivo analysis (virus genotype and phenotype analysis)

Latent occult virus with examples? (3)

The pathogenic viruses having the ability to lie dormant (or latent) within host cell; called lysogenic part of lifecycle.

- Enveloped viruses
- Paramyxoviruses, Retroviruses, Arena-virus
- No disruption of cellular functions
- Promotes shredding of viral Antigens from the cell surface

Examples: Three examples;

i. HSV
ii. Adenovirus
iii. SSPE

Replication of retroviruses? (5)

Retrovirus replication pattern:

Once inside the host cell cytoplasm, the virus uses its own RT enzyme to produce DNA from its RNA genome. The reverse of the usual pattern, thus retro (backwards) The new DNA is then incorporated into the host cell genome by an integrase enzyme referred to as a provirus.The host cell then treats the viral DNA as part of its own genome, transcribing and translating the viral genes along with the cell's own genes. The proteins are produced which are required to assemble new copies of the virus. It is difficult to detect the virus until it has infected the host. Infection will persist indefinitely.

The gag gene --- capsid protein – The pol gene --- molecules of reverse

Transcriptase – The enveloped gene --- molecules of the envelope protein.

What are the types of pathogenesis in interaction of virus and host cells?

- Inapparent
- Apparent
- Acute
- Subacute
- Chronic
- Latency

What is fluorescence?

The ability of substances to absorb short wavelengths of light (ultraviolet) and give off light at a longer wavelength (visible) is known as fluorescence.

What is gene?

Gene is a segment of DNA molecule that encodes for synthesis of a single polypeptide.

What is cell culture?

Cell culture was first successfully undertaken by Ross Harrison in 1907. Cell culture is the process cells which cells are grown under controlled conditions, generally outside their natural environment. After the cells of interest have been isolated from living tissue, they can subsequently be maintained under carefully controlled conditions.

Subfamilies of herpiviridae?

Alphaherpesvirinae

- Herpes Simplex Virus group called Alpha Herpesvirinae
- Latent infections in sensory ganglion, genome size 120-180kbp
- Simplexvirus, Varicellovirus, Human herpes virus-I and II, Human herpes virus-III.

Betaherpesvirinae

- Cytomegalovirus group called Beta Herpesvirinae
- Restricted host range, genome size 140-253 kbp
- Cytomegalovirus, Muromegalovirus, Roseolovirus, Human herpes Virus-V, Mouse cytomegalovirus-I, Human herpes virus-VI and VII.

Gammaherpesvrinae

- Lymphnproliferative Virus group called Gamma Herpesvirinae.
- Infection of lymphoblastic cells, genome size 105-175 kbp
- Lymphocryptovirus, Rhadinovirus, Human herpesvirus-IV, Human Herpes virus-VIII.

Differentiate between DNA dependent RNA polymerase and RNA dependent RNA polymerase.

DNA dependent RNA polymerase

- DNA polymerase is the enzyme which synthesizes new DNA molecules from DNA nucleotides in a process called DNA replication.
- DNA dependent DNA polymerases use in DNA replication to form a double stranded DNA strand.
- DNA polymerase requires a primer for the initiation of replication.
- DNA polymerase possesses exo-nuclease activity.
- DNA polymerase synthesizes the entire chromosome.

RNA dependent RNA polymerase

- RNA polymerase is the enzyme responsible for the synthesis of RNA molecules from DNA in a process called transcription.
- RNA dependent RNA polymerase use in transcription to convert RNA into mRNA
- RNA polymerase does not require a primer for the initiation of transcription.
- RNA polymerase lacks exonuclease activity.
- RNA polymerase stops the synthesis at the transcription termination site.

Write the principle of scanned probe microscope.

PRINCIPLE:

The interaction between the sharp probe and surface provides 3D topographic image of surface at the atomic scale. Two most popular modes:

1. Scanning Tunneling Microscopy (STM)

2. Atomic Force Microscopy (AFM)

Difference between introns and exons?

Intron: Non-coding regions on viral genome.

Exon: coding region on viral genome is exon.

What is the difference between capsid and envelope?

Capsid

- Virus particles contain the viral genome packaged in a protein coat called the capsid.

- In some viruses, the capsid is covered by an envelope comprise of lipid, proteins or carbohydrates bilayer.

Envelope

- Mostly host specific derived from the host cell membranes.
- Prevent digestion by cellular enzymes helps virus to enter host.

Note on Herpes virus? 3

Herpes virus is a type of virus that causes herpes infections and has DNA as its genetic material. The herpes simplex virus, also known as HSV, causes herpes. Herpes can appear in various parts of the body, most commonly on the genitals or mouth. There are two types of the herpes simplex virus:

HSV-1: primarily causes oral herpes, and is generally responsible for cold sores and fever blisters around the mouth and on the face.

HSV-2: primarily causes genital herpes, and is generally responsible for genital herpes outbreaks.

Baltimore classification seven classes. 5

Baltimore classification:

David Baltimore 1971

7 CLASSES:

- GROUP I: double stranded DNA viruses (Adeno, Herpes)
- GROUPII: single stranded DNA viruses (Parvo)
- GROUP III: double stranded RNA viruses (Reo)
- GROUP IV: positive sense single stranded RNA viruses (Polio, Picorna)
- GROUP V: negative sense single stranded RNA viruses (ortho)
- GROUP VI: Single stranded RNA viruses (Retero)
- GROUP VII: Double stranded DNA viruses (Hepadna)

Characteristics of viruses?

The virus particle called virions and consist of nucleic acid surrounded by protein coat.

Capsid Shapes:

- Isometric
- Helical
- Complex

Two types of virions:

- Naked – without envelope
- Enveloped –surrounded by lipid membrane

Steps in animal virus infection cycle? 5

- Viral replication involves six steps: attachment, penetration, uncoating, replication, assembly and release.
- During attachment and penetration, the virus attaches itself to a host cell and injects its genetic material into it.
- During uncoating, replication, and assembly, the viral DNA or RNA incorporates itself into the host cell's genetic material and induces it to replicate the viral genome.
- During release, the newly-created viruses are released from the host cell, either by causing the cell to break apart, waiting for the cell to die, or by budding off through the cell membrane.

Name the receptors of rabies, influenza virus?

The acetylcholine receptor serves as a cellular receptor for rabies virus.

What is bacteriophage and explain the lytic and lysogenic cycle? 3

A bacteriophage also known informally as a phage is a virus that infects and replicates within bacteria and archaea.

Lytic cycle: take over the machinery of the cell to make phage components. They then destroy or lyse, the cell, releasing new phage particles. Lysogenic phages incorporate their nucleic acids into the chromosome of the host cell and replicate with it as a unit without destroying the cell.

Lysogenic cycle: The phage infects a bacterium and inserts its DNA into the bacterial chromosome, allowing the phage DNA (now called a prophage) to be copied and passed on along with the cell's own DNA.

Write name 2 types of scanned probe microscopy? 3

Two most popular modes:

1. Scanning Tunnelling Microscopy (STM)
2. Atomic Force Microscopy (AFM)

Uses of immunofluorescence method? 3

Immunofluorescence techniques are increasingly used as a diagnostic tool in clinical immunology and pathology because of their flexibility and specificity. The main applications at present are detection of circulating autoantibodies, detection of antigens in cells and cell membranes, detection of immune complexes in biopsy material and detection of circulating immune cells.

Reason behind the spontaneous mutation?

Mechanism of genome replication supports the spontaneous mutation. RNA dependent RNA polymerase shows higher error rate than DNA dependent DNA polymerase.

Replication cycle of retrovirus? 5

Early steps

- From the cell surface to the nucleus
- Retroviral particles must bind specifically to their target cells
- Cross the plasma membrane
- Reverse-transcribe their RNA genome
- Uncoat the cores
- Find their way to the nuclear membrane
- Penetrate into the nucleus and hijack the cellular machinery

LATER STAGE

- Transcription
- Transduction
- Assembly
- Maturation
- Budding
- Release

What is life?

A complex set of processes resulting from the actions of proteins specified by nucleic acids. (Nucleic acids of living things are in action all the time).

Three genetic techniques briefly explain.

In-vitro analysis:

- Isolation of nucleic acids from virus preparations
- Direct analysis of DNA using restriction endonucleases enzyme
- Use of vectors in DNA sequencing
- Electron microscopy
- RNA sequence analysis using Reverse transcriptase

Molecular analysis:

- Molecular cloning
- Gel electrophoresis
- Hybridization of Complementary nucleotide sequences

Phenotypic analysis

- Standard technique in virology
- Cell-free extracts to translate mRNAs
- Transfection tests
- In-vitro protein formation
- In-vivo protein replication.